

Erez Yosef Urbach

Curriculum Vitae

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Education

- 2026 – **Postdoctoral member**, *University of Amsterdam*, Amsterdam
- 2024 – 2026 **Postdoctoral member**, *Institute for Advanced Study*, Princeton NJ
- 2024 **Postdoctoral position**, *Weizmann Institute of Science*, Rehovot
- 2019–2024 **PhD in Theoretical Physics**, *Weizmann Institute of Science*, Rehovot
Thesis subject: Holography, Quantum Gravity and String Stars
Advisors: Prof. Ofer Aharony and Prof. Micha Berkooz
- 2016–2019 **MSc in Theoretical Physics**, *Weizmann Institute of Science*, Rehovot
Experimental project with Prof. Efi Efrati.
Thesis subject: Generalized Hawking-Page transitions
Advisors: Prof. Ofer Aharony and Prof. Micha Berkooz
- 2013–2016 **BSc in Physics and Mathematics**, *The Hebrew University*, Jerusalem
Final grade: 96.25 (Magna cum laude).
“Amirim” (science faculty honors program) project with Prof. Barak Kol.

Talks in Scientific Conferences

- 2025 **Holography@Delta Institute for Theoretical Physics**, *University of Amsterdam*
A spin on Hagedorn temperature and string stars
- 2023 **Conformal Bootstrap, CFTs, and Gravity**, *Technion*
The black hole/string transition in anti-de Sitter space
- 2023 **First DIP Conference on Holography and the Swampland**, *Ben-Gurion University of the Negev*
The black hole/string transition in anti-de Sitter space
- 2022 **12th Crete Regional Meeting in String Theory**
String stars in anti de Sitter space
- 2022 **Joint Seminar in Theoretical High Energy Physics**, *Neve Shalom*
String stars in anti de Sitter space
- 2022 **IPS Conference 2022 - Parallel Session**
String stars in anti de Sitter space
- 2021 **Strings 2021 gong-show**, *São Paulo*, Brazil
The entanglement entropy of typical pure states and replica wormholes
- 2021 **IPS Conference 2021 - Parallel Session**
A Derivation of AdS/CFT for Vector Models

2018 **APS March Meeting 2018 - Parallel Session**, Los Angeles
How the Hopper Pops: Unraveling Visco-elastic Instabilities Through a Metric Description

Publications

E.Y. Urbach, *String stars, small black holes, and the supersymmetric index*, 2608.21511.

J. Seitz and E.Y. Urbach, *A spin on Hagedorn temperatures and string stars*, *JHEP* **06** (2026) 245 [2510.17951].

A. Blommaert, J. Kudler-Flam and E.Y. Urbach, *Absolute entropy and the observer's no-boundary state*, *JHEP* **11** (2025) 113 [2505.14771].

O. Aharony and E.Y. Urbach, *Type II string theory on $AdS_3 \times S^3 \times T^4$ and symmetric orbifolds*, *Phys. Rev. D* **110** (2024) 046028 [2406.14605].

E.Y. Urbach, *The black hole/string transition in AdS_3 and confining backgrounds*, *JHEP* **09** (2023) 156 [2303.09567].

O. Aharony, S.M. Chester, T. Sheaffer and E.Y. Urbach, *Explicit holography for vector models at finite N , volume and temperature*, *JHEP* **03** (2023) 016 [2208.13607].

M. Berkooz, A. Sharon, N. Silberstein and E.Y. Urbach, *Onset of Quantum Chaos in Random Field Theories*, *Phys. Rev. Lett.* **129** (2022) 071601 [2207.11980].

E.Y. Urbach, *String stars in anti de Sitter space*, *JHEP* **04** (2022) 072 [2202.06966].

M. Berkooz, A. Sharon, N. Silberstein and E.Y. Urbach, *Onset of quantum chaos in disordered CFTs*, *Phys. Rev. D* **106** (2022) 045007 [2111.06108].

O. Aharony, S.M. Chester and E.Y. Urbach, *AdS from CFT for scalar QED*, *Phys. Rev. D* **104** (2021) 126011 [2109.05512].

E.Y. Urbach, *The entanglement entropy of typical pure states and replica wormholes*, *JHEP* **12** (2021) 125 [2105.15059].

O. Aharony, S.M. Chester and E.Y. Urbach, *A Derivation of AdS/CFT for Vector Models*, *JHEP* **03** (2021) 208 [2011.06328].

O. Aharony, E.Y. Urbach and M. Weiss, *Generalized Hawking-Page transitions*, *JHEP* **08** (2019) 018 [1904.07502].

Service to the Scientific Community

2023-present **Referee**, *Journal of High Energy Physics (JHEP)*

Teaching Experience

- 2020–current **Teaching Assistant for Quantum Mechanics 2 course**, *Weizmann Institute of Science*
- 2014–2016 **Physics Teaching Assistant in “The Virtual Campus” project**, *The Center for Educational Technology*

Awards and Achievements

- 2024–2025 **Rothschild Fellow**, *Yad Hanadiv*
- 2023 **The John F. Kennedy Prize for academic excellence and scientific accomplishments**, *Weizmann Institute of Science*
- 2014, 2015 **Full scholarship for academic excellence (“Amirim program”)**, *The Hebrew University*
- 2013, 2014 **Outstanding students Dean’s list**, *The Hebrew University*
- 2012 **Second Prize**, *“Kadima Mada” Science Projects Competition*
- 2011 **Honorable Mention**, *ACM Special Award on Intel Science Fair (ISEF)*
- 2011 **First Prize**, *Israel Young Scientists and Engineering Competition*